# Problem-01:

Consider a direct mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find-

1. Number of bits in tag
2. Tag directory size

# Solution-

Given-

- Cache memory size = 16 KB
- Block size = Frame size = Line size = 256 bytes
- Main memory size = 128 KB

We consider that the memory is byte addressable.

## Number of Bits in Physical Address-

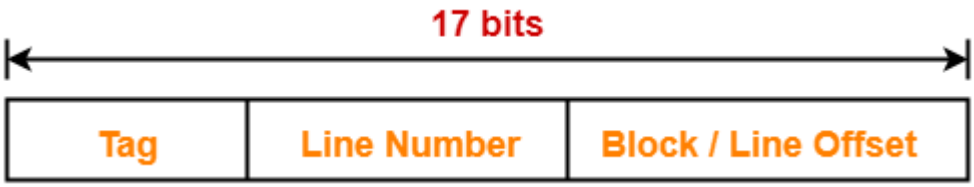
We have,

Size of main memory

= 128 KB

= $2^{17}$ bytes

Thus, Number of bits in physical address = 17 bits




## Number of Bits in Block Offset-

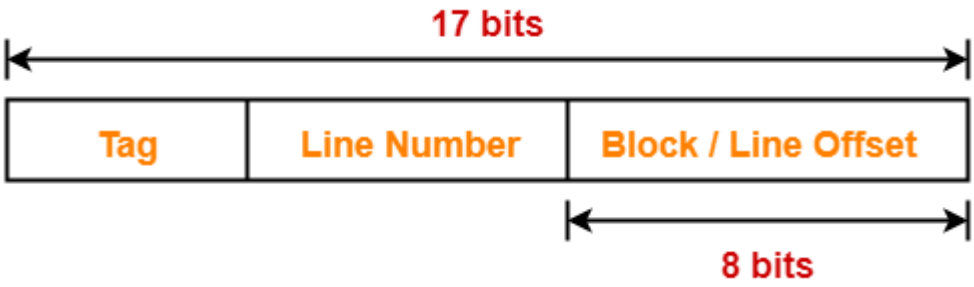
We have,

Block size

= 256 bytes

= $2^8$ bytes

Thus, Number of bits in block offset = 8 bits




## Number of Bits in Line Number-

Total number of lines in cache

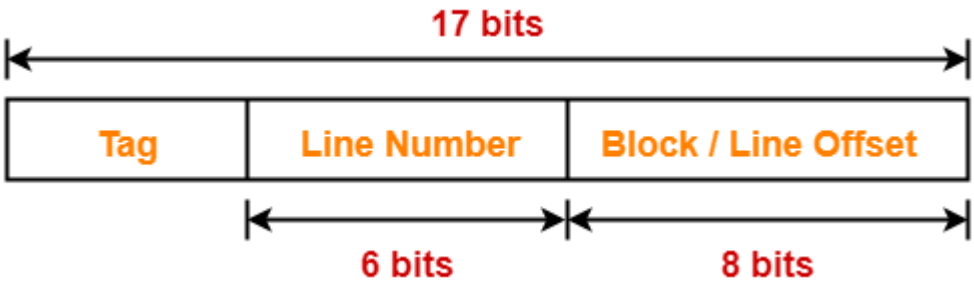
= Cache size / Line size

= 16 KB / 256 bytes

= $2^{14}$ bytes / $2^8$ bytes

= $2^6$ lines

Thus, Number of bits in line number = 6 bits




## Number of Bits in Tag-

Number of bits in tag

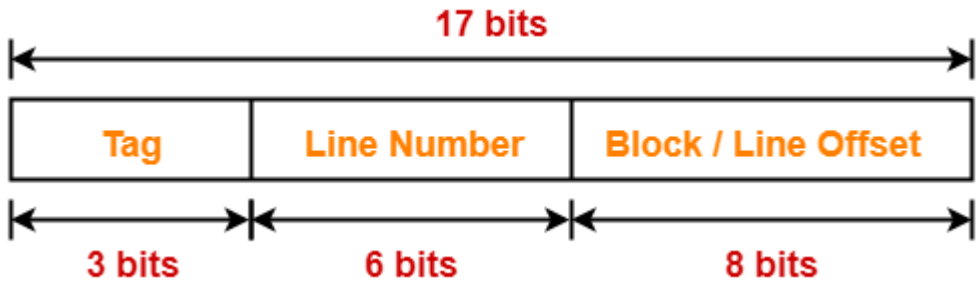
= Number of bits in physical address – (Number of bits in line number + Number of bits in block offset)

= 17 bits – (6 bits + 8 bits)

= 17 bits – 14 bits

= 3 bits

Thus, Number of bits in tag = 3 bits




## Tag Directory Size-

Tag directory size

= Number of tags x Tag size

= Number of lines in cache x Number of bits in tag

= $2^6$ x 3 bits

= 192 bits

= 24 bytes

Thus, size of tag directory = 24 bytes

# Problem-02:

Consider a direct mapped cache of size 512 KB with block size 1 KB. There are 7 bits in the tag. Find-

1. Size of main memory
2. Tag directory size

# Solution-

Given-

- Cache memory size = 512 KB
- Block size = Frame size = Line size = 1 KB
- Number of bits in tag = 7 bits

We consider that the memory is byte addressable.

## Number of Bits in Block Offset-

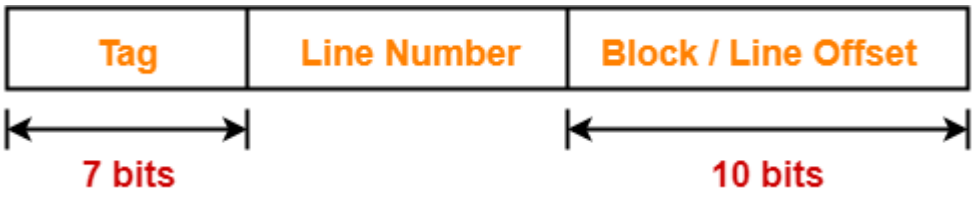
We have,

Block size

= 1 KB

= $2^{10}$ bytes

Thus, Number of bits in block offset = 10 bits




## Number of Bits in Line Number-

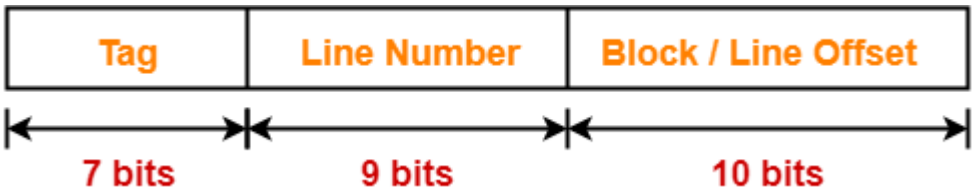
Total number of lines in cache

= Cache size / Line size

= 512 KB / 1 KB

= $2^{9}$ lines

Thus, Number of bits in line number = 9 bits




## Number of Bits in Physical Address-

Number of bits in physical address

= Number of bits in tag + Number of bits in line number + Number of bits in block offset

= 7 bits + 9 bits + 10 bits

= 26 bits

Thus, Number of bits in physical address = 26 bits

## Size of Main Memory-

We have,

Number of bits in physical address = 26 bits

Thus, Size of main memory

= $2^{26}$ bytes

= 64 MB

## Tag Directory Size-

Tag directory size

= Number of tags x Tag size

= Number of lines in cache x Number of bits in tag

= $2^{9}$ x 7 bits

= 3584 bits

= 448 bytes

Thus, size of tag directory = 448 bytes

# Problem-03:

Consider a direct mapped cache with block size 4 KB. The size of main memory is 16 GB and there are 10 bits in the tag. Find-

1. Size of cache memory
2. Tag directory size

# Solution-

Given-

- Block size = Frame size = Line size = 4 KB
- Size of main memory = 16 GB
- Number of bits in tag = 10 bits

We consider that the memory is byte addressable.

## Number of Bits in Physical Address-

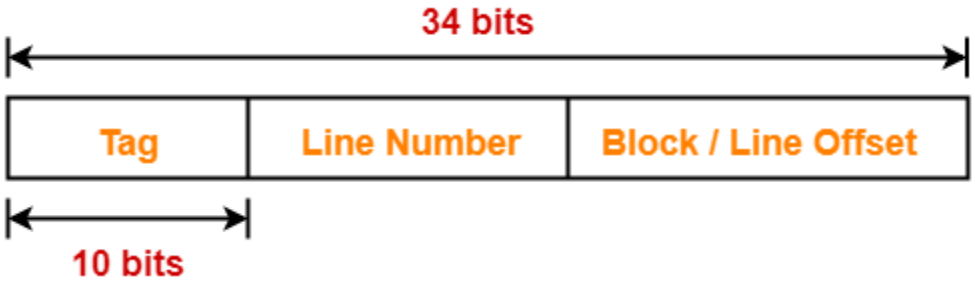
We have,

Size of main memory

= 16 GB

= $2^{34}$ bytes

Thus, Number of bits in physical address = 34 bits



## Number of Bits in Block Offset-

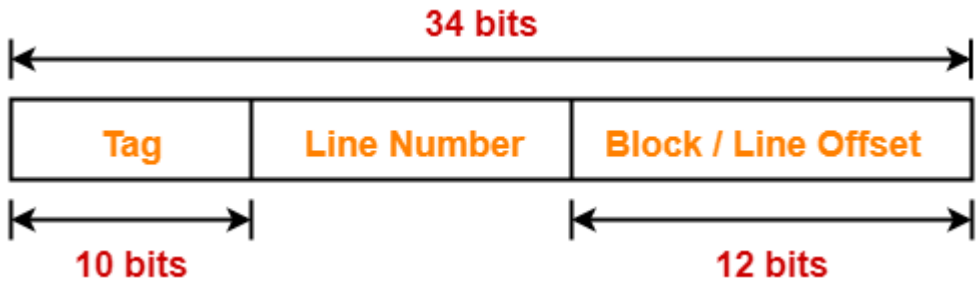
We have,

Block size

= 4 KB

= $2^{12}$ bytes

Thus, Number of bits in block offset = 12 bits




## Number of Bits in Line Number-

Number of bits in line number

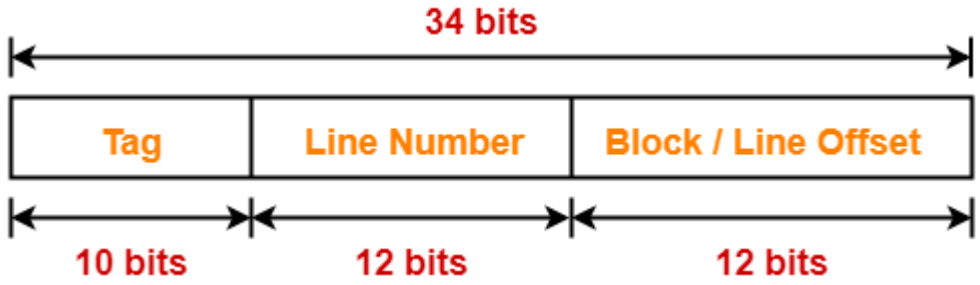
= Number of bits in physical address – (Number of bits in tag + Number of bits in block offset)

= 34 bits – (10 bits + 12 bits)

= 34 bits – 22 bits

= 12 bits

Thus, Number of bits in line number = 12 bits



## Number of Lines in Cache-

We have-

Number of bits in line number = 12 bits

Thus, Total number of lines in cache = $2^{12}$ lines

## Size of Cache Memory-

Size of cache memory

= Total number of lines in cache x Line size

= $2^{12}$ x 4 KB

= $2^{14}$ KB

= 16 MB

Thus, Size of cache memory = 16 MB

## Tag Directory Size-

Tag directory size

= Number of tags x Tag size

= Number of lines in cache x Number of bits in tag

= $2^{12}$ x 10 bits

= 40960 bits

= 5120 bytes

Thus, size of tag directory = 5120 bytes

**Also Read- Practice Problems On Set Associative Mapping**

# Problem-04:

Consider a direct mapped cache of size 32 KB with block size 32 bytes. The CPU generates 32 bit addresses. The number of bits needed for cache indexing and the number of tag bits are respectively-

1. 10, 17
2. 10, 22
3. 15, 17
4. 5, 17

## Solution-

Given-

- Cache memory size = 32 KB
- Block size = Frame size = Line size = 32 bytes
- Number of bits in physical address = 32 bits

### Number of Bits in Block Offset-

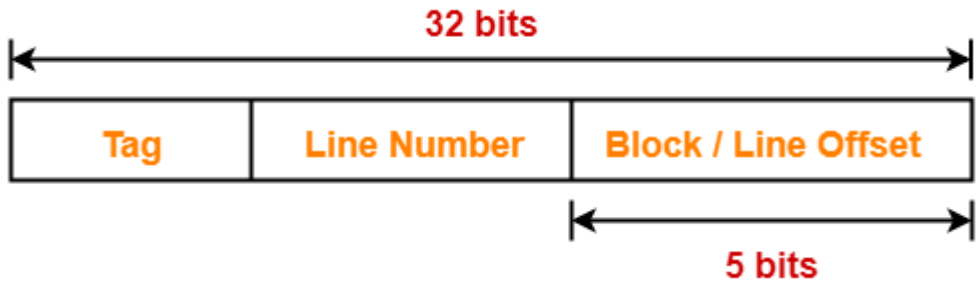
We have,

Block size

= 32 bytes

= $2^5$ bytes

Thus, Number of bits in block offset = 5 bits




### Number of Bits in Line Number-

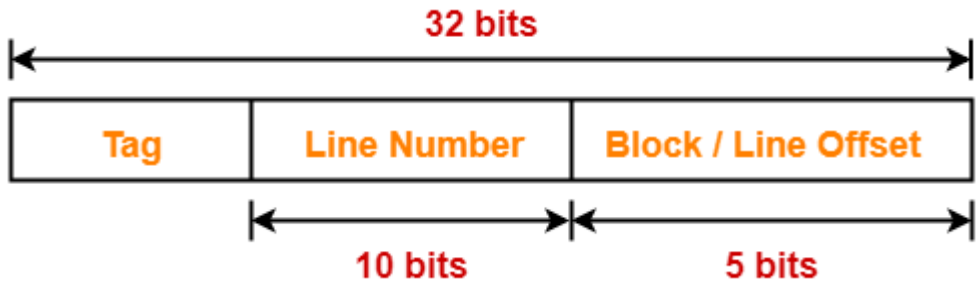
Total number of lines in cache

= Cache size / Line size

= 32 KB / 32 bytes

= $2^{10}$ lines

Thus, Number of bits in line number = 10 bits




## Number of Bits Required For Cache Indexing-

Number of bits required for cache indexing

= Number of bits in line number

= 10 bits

## Number Of Bits in Tag-

Number of bits in tag

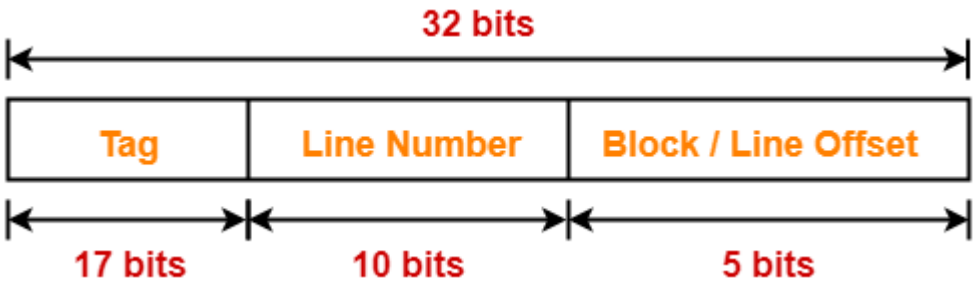
= Number of bits in physical address – (Number of bits in line number + Number of bits in block offset)

= 32 bits – (10 bits + 5 bits)

= 32 bits – 15 bits

= 17 bits

Thus, Number of bits in tag = 17 bits




Thus, Option (A) is correct.

## Problem-05:

Consider a machine with a byte addressable main memory of $2^{32}$ bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

## Solution-

Given-

- Main memory size = $2^{32}$ bytes
- Block size = Frame size = Line size = 32 bytes
- Number of lines in cache = 512 lines

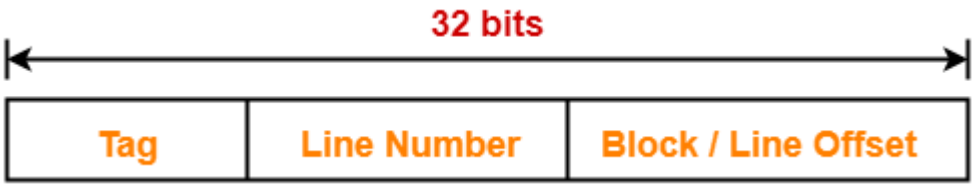
### Number of Bits in Physical Address-

We have,

Size of main memory

= $2^{32}$ bytes

Thus, Number of bits in physical address = 32 bits




## Number of Bits in Block Offset-

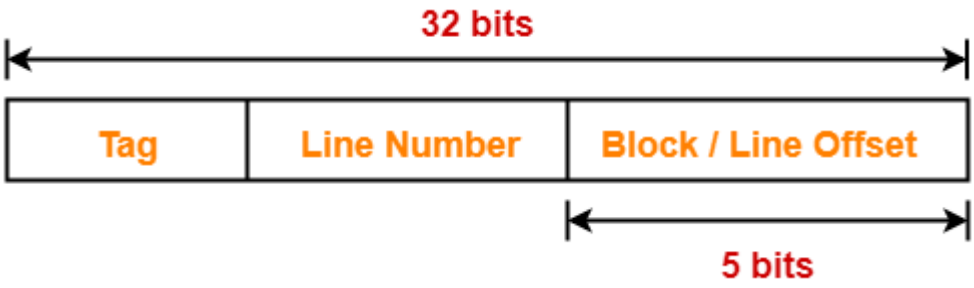
We have,

Block size

= 32 bytes

= $2^5$ bytes

Thus, Number of bits in block offset = 5 bits



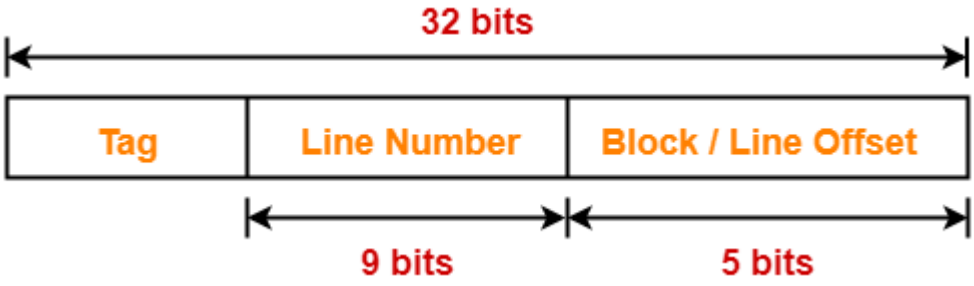

## Number of Bits in Line Number-

Total number of lines in cache

= 512 lines

= $2^9$ lines

Thus, Number of bits in line number = 9 bits




## Number Of Bits in Tag-

Number of bits in tag

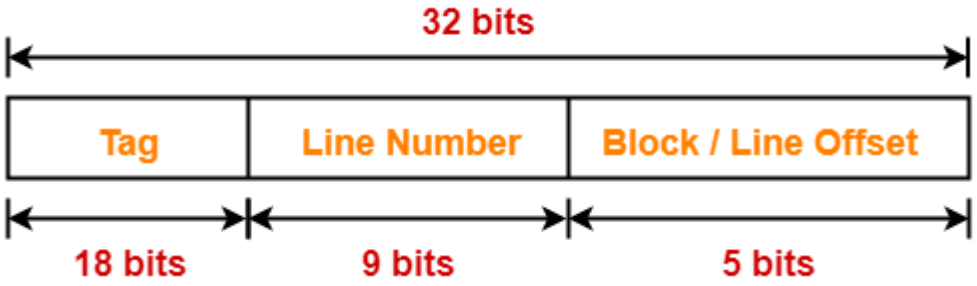
= Number of bits in physical address – (Number of bits in line number + Number of bits in block offset)

= 32 bits – (9 bits + 5 bits)

= 32 bits – 14 bits

= 18 bits

Thus, Number of bits in tag = 18 bits




# Problem-06:

An 8 KB direct-mapped write back cache is organized as multiple blocks, each of size 32 bytes. The processor generates 32 bit addresses. The cache controller maintains the tag information for each cache block comprising of the following-

- 1 valid bit
- 1 modified bit
- As many bits as the minimum needed to identify the memory block mapped in the cache

What is the total size of memory needed at the cache controller to store meta data (tags) for the cache?

1. 4864 bits
2. 6144 bits
3. 6656 bits
4. 5376 bits

# Solution-

Given-

- Cache memory size = 8 KB
- Block size = Frame size = Line size = 32 bytes
- Number of bits in physical address = 32 bits

## Number of Bits in Block Offset-

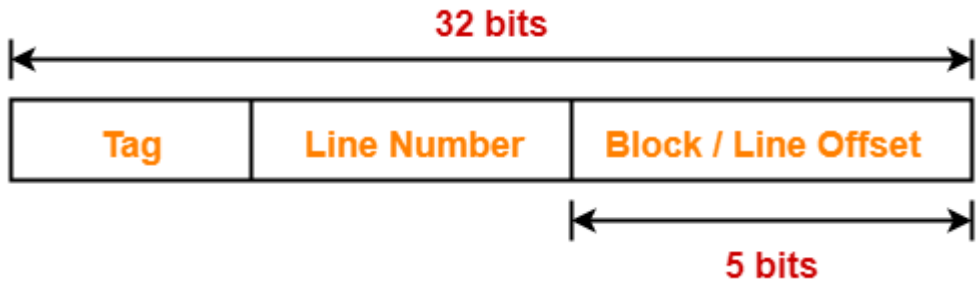
We have,

Block size

= 32 bytes

= $2^5$ bytes

Thus, Number of bits in block offset = 5 bits




## Number of Bits in Line Number-

Total number of lines in cache

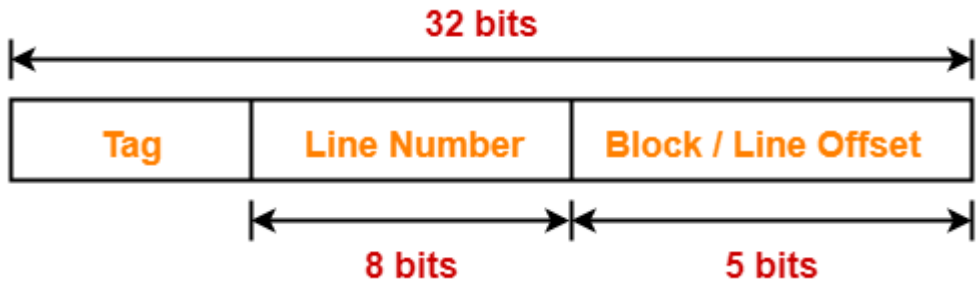
= Cache memory size / Line size

= 8 KB / 32 bytes

= $2^{13}$ bytes / $2^{5}$ bytes

= $2^{8}$ lines

Thus, Number of bits in line number = 8 bits




## Number Of Bits in Tag-

Number of bits in tag

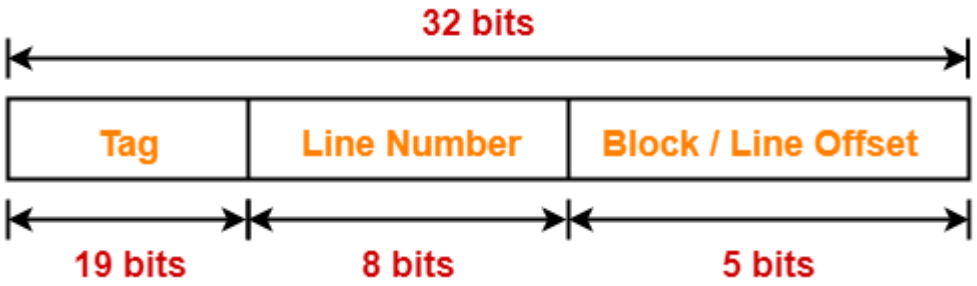
= Number of bits in physical address – (Number of bits in line number + Number of bits in block offset)

= 32 bits – (8 bits + 5 bits)

= 32 bits – 13 bits

= 19 bits

Thus, Number of bits in tag = 19 bits




## Memory Size Needed At Cache Controller-

Size of memory needed at cache controller

= Number of lines in cache x (1 valid bit + 1 modified bit + 19 bits to identify block)

= $2^8$ x 21 bits

= 5376 bits

# PRACTICE PROBLEMS BASED ON FULLY ASSOCIATIVE MAPPING-

## Problem-01:

Consider a fully associative mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find-

1. Number of bits in tag
2. Tag directory size

## Solution-

Given-

- Cache memory size = 16 KB
- Block size = Frame size = Line size = 256 bytes
- Main memory size = 128 KB

We consider that the memory is byte addressable.

### Number of Bits in Physical Address-

We have,

Size of main memory

= 128 KB

= $2^{17}$ bytes

Thus, Number of bits in physical address = 17 bits


## Number of Bits in Block Offset-

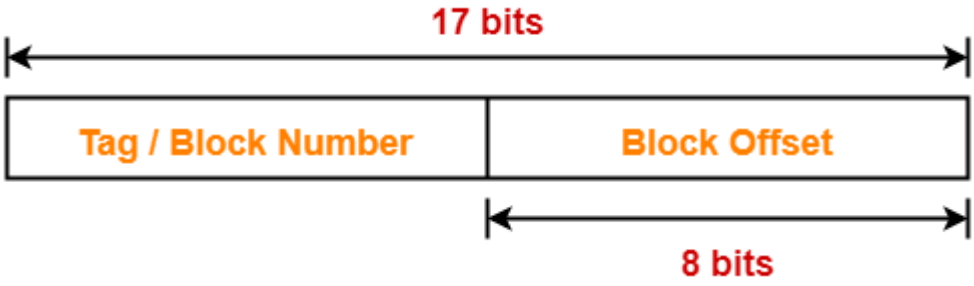
We have,

Block size

= 256 bytes

= $2^8$ bytes

Thus, Number of bits in block offset = 8 bits




## Number of Bits in Tag-

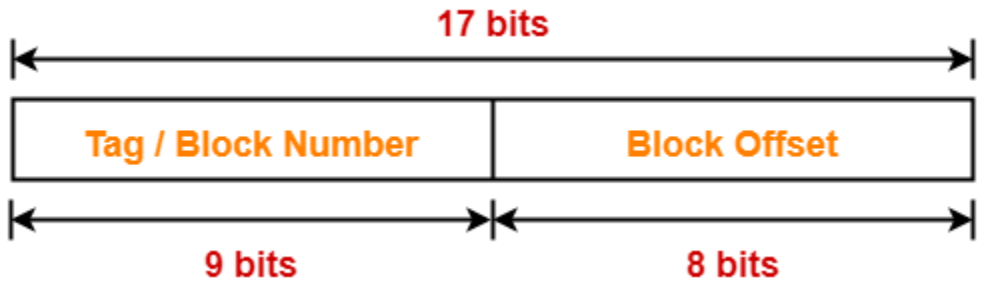
Number of bits in tag

= Number of bits in physical address – Number of bits in block offset

= 17 bits – 8 bits

= 9 bits

Thus, Number of bits in tag = 9 bits




## Number of Lines in Cache-

Total number of lines in cache

= Cache size / Line size

= 16 KB / 256 bytes

= $2^{14}$ bytes / $2^{8}$ bytes

= $2^{6}$ lines

## Tag Directory Size-

Tag directory size

= Number of tags x Tag size

= Number of lines in cache x Number of bits in tag

= $2^{6}$ x 9 bits

= 576 bits

= 72 bytes

Thus, size of tag directory = 72 bytes

# Problem-02:

Consider a fully associative mapped cache of size 512 KB with block size 1 KB. There are 17 bits in the tag. Find-

1. Size of main memory
2. Tag directory size

# Solution-

Given-

- Cache memory size = 512 KB
- Block size = Frame size = Line size = 1 KB
- Number of bits in tag = 17 bits

We consider that the memory is byte addressable.

## Number of Bits in Block Offset-

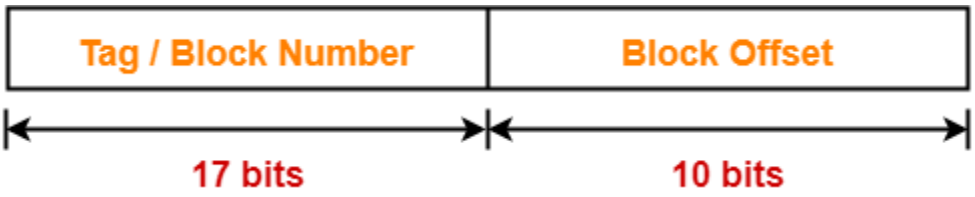
We have,

Block size

= 1 KB

= $2^{10}$ bytes

Thus, Number of bits in block offset = 10 bits




## Number of Bits in Physical Address-

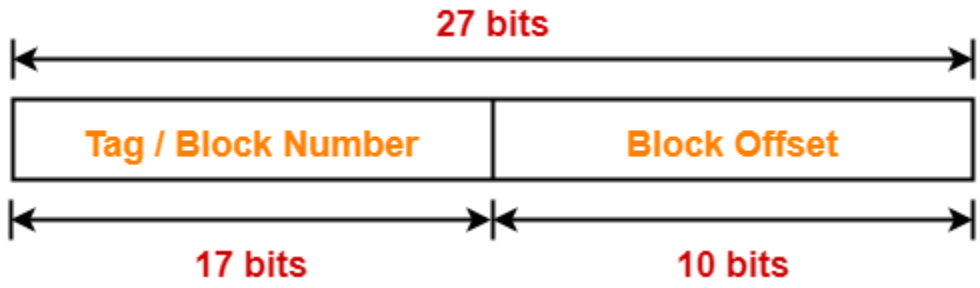
Number of bits in physical address

= Number of bits in tag + Number of bits in block offset

= 17 bits + 10 bits

= 27 bits

Thus, Number of bits in physical address = 27 bits




## Size of Main Memory-

We have,

Number of bits in physical address = 27 bits

Thus, Size of main memory

= $2^{27}$ bytes

= 128 MB

## Number of Lines in Cache-

Total number of lines in cache

= Cache size / Line size

= 512 KB / 1 KB

= 512 lines

= $2^{9}$ lines

## Tag Directory Size-

Tag directory size

= Number of tags x Tag size

= Number of lines in cache x Number of bits in tag

= $2^{9}$ x 17 bits

= 8704 bits

= 1088 bytes

Thus, size of tag directory = 1088 bytes

**Also Read- Practice Problems On Set Associative Mapping**

# Problem-03:

Consider a fully associative mapped cache with block size 4 KB. The size of main memory is 16 GB. Find the number of bits in tag.

# Solution-

Given-

- Block size = Frame size = Line size = 4 KB
- Size of main memory = 16 GB

We consider that the memory is byte addressable.

## Number of Bits in Physical Address-

We have,

Size of main memory

= 16 GB

= $2^{34}$ bytes

Thus, Number of bits in physical address = 34 bits

## Number of Bits in Block Offset-

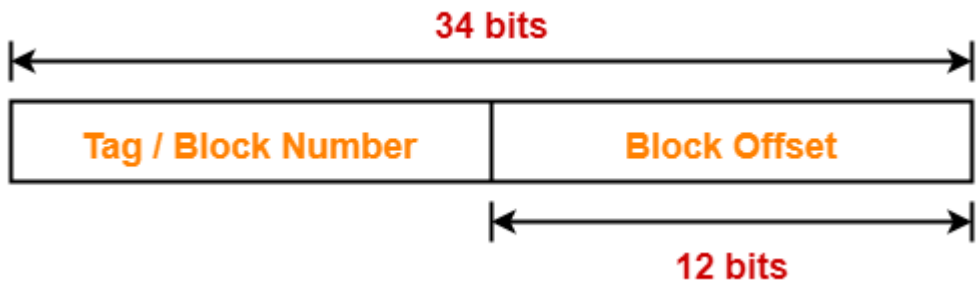
We have,

Block size

= 4 KB

= $2^{12}$ bytes

Thus, Number of bits in block offset = 12 bits




## Number of Bits in Tag-

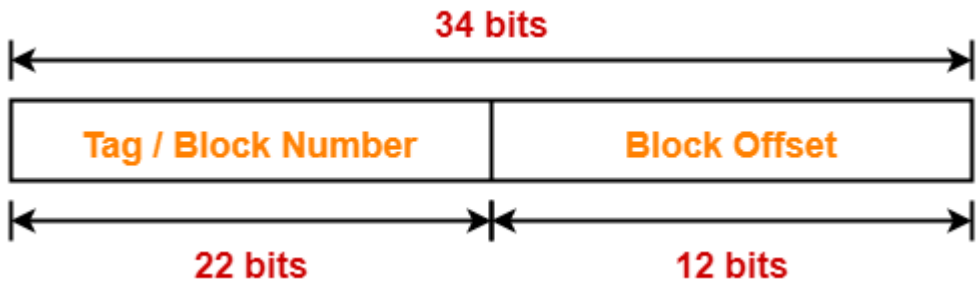
Number of bits in tag

= Number of bits in physical address – Number of bits in block offset

= 34 bits – 12 bits

= 22 bits

Thus, Number of bits in tag = 22 bits



# PRACTICE PROBLEMS BASED ON CACHE MAPPING TECHNIQUES-

## Problem-01:

The main memory of a computer has 2 cm blocks while the cache has 2c blocks. If the cache uses the set associative mapping scheme with 2 blocks per set, then block k of the main memory maps to the set-

1. (k mod m) of the cache
2. (k mod c) of the cache
3. (k mod 2 c) of the cache
4. (k mod 2 cm) of the cache

## Solution-

Given-

- Number of blocks in main memory = 2 cm
- Number of blocks in cache = 2 c
- Number of blocks in one set of cache = 2

### Number of Sets in Cache-

Number of sets in cache

= Number of blocks in cache / Number of blocks in one set

= 2 c / 2

= c

### Required Mapping-

In set associative mapping,

- Block ‘j’ of main memory maps to set number (j modulo number of sets in cache) of the cache.
- So, block ‘k’ of main memory maps to set number (k mod c) of the cache.
- Thus, option (B) is correct.

**Also Read- Practice Problems On Direct Mapping**

# Problem-02:

In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 on wards. The main memory block numbered 'j' must be mapped to any one of the cache lines from-

1. (j mod v) x k to (j mod v) x k + (k – 1)
2. (j mod v) to (j mod v) + (k – 1)
3. (j mod k) to (j mod k) + (v – 1)
4. (j mod k) x v to (j mod k) x v + (v – 1)

# Solution-

Let-

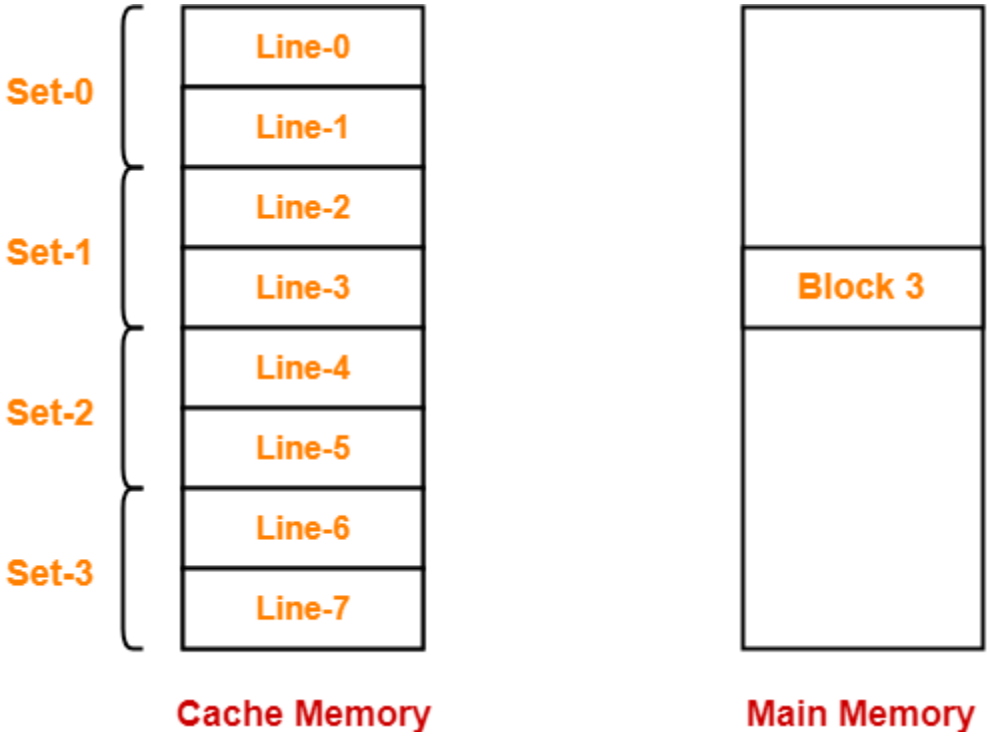
- 2-way set associative mapping is used, then k = 2
- Number of sets in cache = 4, then v = 4
- Block number '3' has to be mapped, then j = 3

The given options on substituting these values reduce to-

1. (3 mod 4) x 2 to (3 mod 4) x 2 + (2 – 1) = 6 to 7
2. (3 mod 4) to (3 mod 4) + (2 – 1) = 3 to 4
3. (3 mod 2) to (3 mod 2) + (4 – 1) = 1 to 4
4. (3 mod 2) x 4 to (3 mod 2) x 4 + (4 – 1) = 4 to 7

Now, we have been asked to what range of cache lines, block number 3 can be mapped.

According to the above data, the cache memory and main memory will look like-




In set associative mapping,

- Block ‘j’ of main memory maps to set number (j modulo number of sets in cache) of the cache.
- So, block 3 of main memory maps to set number (3 mod 4) = 3 of cache.
- Within set number 3, block 3 can be mapped to any of the cache lines.
- Thus, block 3 can be mapped to cache lines ranging from 6 to 7.

Thus, Option (A) is correct.

## Problem-03:

A block-set associative cache memory consists of 128 blocks divided into four block sets . The main memory consists of 16,384 blocks and each block contains 256 eight bit words.

1. How many bits are required for addressing the main memory?
2. How many bits are needed to represent the TAG, SET and WORD fields?

## Solution-

Given-

- Number of blocks in cache memory = 128
- Number of blocks in each set of cache = 4
- Main memory size = 16384 blocks
- Block size = 256 bytes
- 1 word = 8 bits = 1 byte

## Main Memory Size-

We have-

Size of main memory

= 16384 blocks

= 16384 x 256 bytes

= $2^{22}$ bytes

Thus, Number of bits required to address main memory = 22 bits

## Number of Bits in Block Offset-

We have-

Block size

= 256 bytes

= $2^{8}$ bytes

Thus, Number of bits in block offset or word = 8 bits

## Number of Bits in Set Number-

Number of sets in cache

= Number of lines in cache / Set size

= 128 blocks / 4 blocks

= 32 sets

= $2^{5}$ sets

Thus, Number of bits in set number = 5 bits

## Number of Bits in Tag Number-

Number of bits in tag

= Number of bits in physical address – (Number of bits in set number + Number of bits in word)

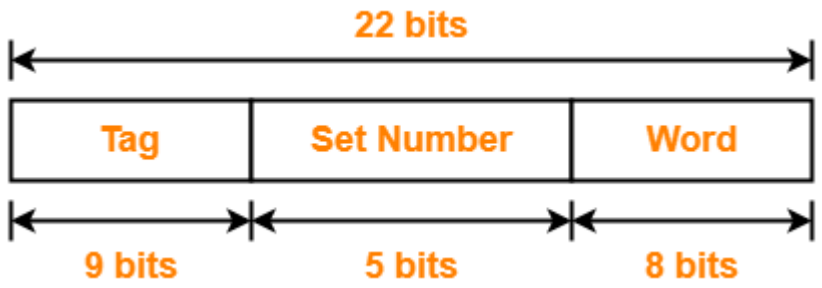
= 22 bits – (5 bits + 8 bits)

= 22 bits – 13 bits

= 9 bits

Thus, Number of bits in tag = 9 bits

Thus, physical address is-




# Problem-04:

A computer has a 256 KB, 4-way set associative, write back data cache with block size of 32 bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

## Part-01:

The number of bits in the tag field of an address is-

1. 11
2. 14
3. 16
4. 27

## Part-02:

The size of the cache tag directory is-

1. 160 Kbits
2. 136 Kbits
3. 40 Kbits
4. 32 Kbits

# Solution-

Given-

- Cache memory size = 256 KB
- Set size = 4 blocks
- Block size = 32 bytes
- Number of bits in physical address = 32 bits

## Number of Bits in Block Offset-

We have-

Block size

= 32 bytes

= $2^5$ bytes

Thus, Number of bits in block offset = 5 bits

## Number of Lines in Cache-

Number of lines in cache

= Cache size / Line size

= 256 KB / 32 bytes

= $2^{18}$ bytes / $2^5$ bytes

= $2^{13}$ lines

Thus, Number of lines in cache = $2^{13}$ lines

## Number of Sets in Cache-

Number of sets in cache

= Number of lines in cache / Set size

= $2^{13}$ lines / $2^{2}$ lines

= $2^{11}$ sets

Thus, Number of bits in set number = 11 bits

## Number of Bits in Tag-

Number of bits in tag

= Number of bits in physical address – (Number of bits in set number + Number of bits in block offset)

= 32 bits – (11 bits + 5 bits)

= 32 bits – 16 bits

= 16 bits

Thus, Number of bits in tag = 16 bits

## Tag Directory Size-

Size of tag directory

= Number of lines in cache x Size of tag

= $2^{13}$ x (16 bits + 2 valid bits + 1 modified bit + 1 replacement bit)

= $2^{13}$ x 20 bits

= 163840 bits

= 20 KB or 160 Kbits

Thus,

- For part-01, Option (C) is correct.
- For part-02, Option (A) is correct.

**Also Read- Practice Problems On Fully Associative Mapping**

# Problem-05:

A 4-way set associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____.

# Solution-

Given-

- Set size = 4 lines
- Cache memory size = 16 KB
- Block size = 8 words
- 1 word = 32 bits = 4 bytes
- Main memory size = 4 GB

## Number of Bits in Physical Address-

We have,

Main memory size

= 4 GB

= $2^{32}$ bytes

Thus, Number of bits in physical address = 32 bits

## Number of Bits in Block Offset-

We have,

Block size

= 8 words

= 8 x 4 bytes

= 32 bytes

= $2^{5}$ bytes

Thus, Number of bits in block offset = 5 bits

## Number of Lines in Cache-

Number of lines in cache

= Cache size / Line size

= 16 KB / 32 bytes

= $2^{14}$ bytes / $2^{5}$ bytes

= $2^{9}$ lines

= 512 lines

Thus, Number of lines in cache = 512 lines

## Number of Sets in Cache-

Number of sets in cache

= Number of lines in cache / Set size

= 512 lines / 4 lines

= $2^{9}$ lines / $2^{2}$ lines

= $2^{7}$ sets

Thus, Number of bits in set number = 7 bits

## Number of Bits in Tag-

Number of bits in tag

= Number of bits in physical address – (Number of bits in set number + Number of bits in block offset)

= 32 bits – (7 bits + 5 bits)

= 32 bits – 12 bits

= 20 bits

Thus, number of bits in tag = 20 bits

# Problem-06:

If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

1. Width of tag comparator
2. Width of set index decoder
3. Width of way selection multiplexer
4. Width of processor to main memory data bus

# Solution-

Since block size is unchanged, so number of bits in block offset will remain unchanged.

## Effect On Width Of Tag Comparator-

- Associativity of cache is doubled means number of lines in one set is doubled.
- Since number of lines in one set is doubled, therefore number of sets reduces to half.
- Since number of sets reduces to half, so number of bits in set number decrements by 1.
- Since number of bits in set number decrements by 1, so number of bits in tag increments by 1.
- Since number of bits in tag increases, therefore width of tag comparator also increases.

## Effect On Width of Set Index Decoder-

- Associativity of cache is doubled means number of lines in one set is doubled.
- Since number of lines in one set is doubled, therefore number of sets reduces to half.
- Since number of sets reduces to half, so number of bits in set number decrements by 1.
- Since number of bits in set number decreases, therefore width of set index decoder also decreases.

## Effect On Width Of Way Selection Multiplexer-

- Associativity of cache (k) is doubled means number of lines in one set is doubled.
- New associativity of cache is 2k.
- To handle new associativity, size of multiplexers must be 2k x 1.
- Therefore, width of way selection multiplexer increases.

## Effect On Width Of Processor To Main Memory Data Bus-

- Processor to main memory data bus has nothing to do with cache associativity.

- It depends on the number of bits in block offset which is unchanged here.
- So, width of processor to main memory data bus is not affected and remains unchanged.

Thus, Option (D) is correct.

## Problem-07:

Consider a direct mapped cache with 8 cache blocks (0-7). If the memory block requests are in the order-

3, 5, 2, 8, 0, 6, 3, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24

Which of the following memory blocks will not be in the cache at the end of the sequence?

1. 3
2. 18
3. 20
4. 30

Also, calculate the hit ratio and miss ratio.

## Solution-

We have,

- There are 8 blocks in cache memory numbered from 0 to 7.
- In direct mapping, a particular block of main memory is mapped to a particular line of cache memory.
- The line number is given by-

Cache line number = Block address modulo Number of lines in cache

For the given sequence-

- Requests for memory blocks are generated one by one.
- The line number of the block is calculated using the above relation.
- Then, the block is placed in that particular line.
- If already there exists another block in that line, then it is replaced.

| | Cache Memory |
|---|---|
| Line-0 | ~~8~~, ~~0~~, ~~16~~, 24 |
| Line-1 | ~~9~~, ~~17~~, ~~25~~, 17 |
| Line-2 | ~~2~~, ~~18~~, ~~2~~, 82 |
| Line-3 | 3 |
| Line-4 | 20 |
| Line-5 | 5 |
| Line-6 | ~~6~~, 30 |
| Line-7 | 63 |

**Blocks requests in order**
**Calculation of line number**

| | |
|---|---|
| 3 % 8 = 3 | (Miss) |
| 5 % 8 = 5 | (Miss) |
| 2 % 8 = 2 | (Miss) |
| 8 % 8 = 0 | (Miss) |
| 0 % 8 = 0 | (Miss) |
| 6 % 8 = 6 | (Miss) |
| 3 % 8 = 3 | (Hit) |
| 9 % 8 = 1 | (Miss) |
| 16 % 8 = 0 | (Miss) |
| 20 % 8 = 4 | (Miss) |
| 17 % 8 = 1 | (Miss) |
| 25 % 8 = 1 | (Miss) |
| 18 % 8 = 2 | (Miss) |
| 30 % 8 = 6 | (Miss) |
| 24 % 8 = 0 | (Miss) |
| 2 % 8 = 2 | (Miss) |
| 63 % 8 = 7 | (Miss) |
| 5 % 8 = 5 | (Hit) |
| 82 % 8 = 2 | (Miss) |
| 17 % 8 = 1 | (Miss) |
| 24 % 8 = 0 | (Hit) |

Thus,

- Out of given options, only block-18 is not present in the main memory.
- Option-(B) is correct.
- Hit ratio = 3 / 20
- Miss ratio = 17 / 20

# Problem-08:

Consider a fully associative cache with 8 cache blocks (0-7). The memory block requests are in the order-

4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7

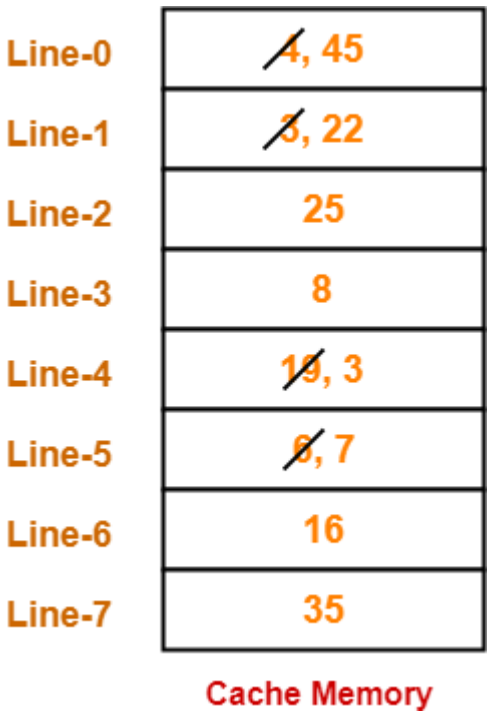
If LRU replacement policy is used, which cache block will have memory block 7?

Also, calculate the hit ratio and miss ratio.

# Solution-

We have,

- There are 8 blocks in cache memory numbered from 0 to 7.
- In fully associative mapping, any block of main memory can be mapped to any line of the cache that is freely available.
- If all the cache lines are already occupied, then a block is replaced in accordance with the replacement policy.




Thus,

- Line-5 contains the block-7.

- Hit ratio = 5 / 17
- Miss ratio = 12 / 17

## Problem-09:

Consider a 4-way set associative mapping with 16 cache blocks. The memory block requests are in the order-

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155

If LRU replacement policy is used, which cache block will not be present in the cache?

1. 3
2. 8
3. 129
4. 216

Also, calculate the hit ratio and miss ratio.

## Solution-

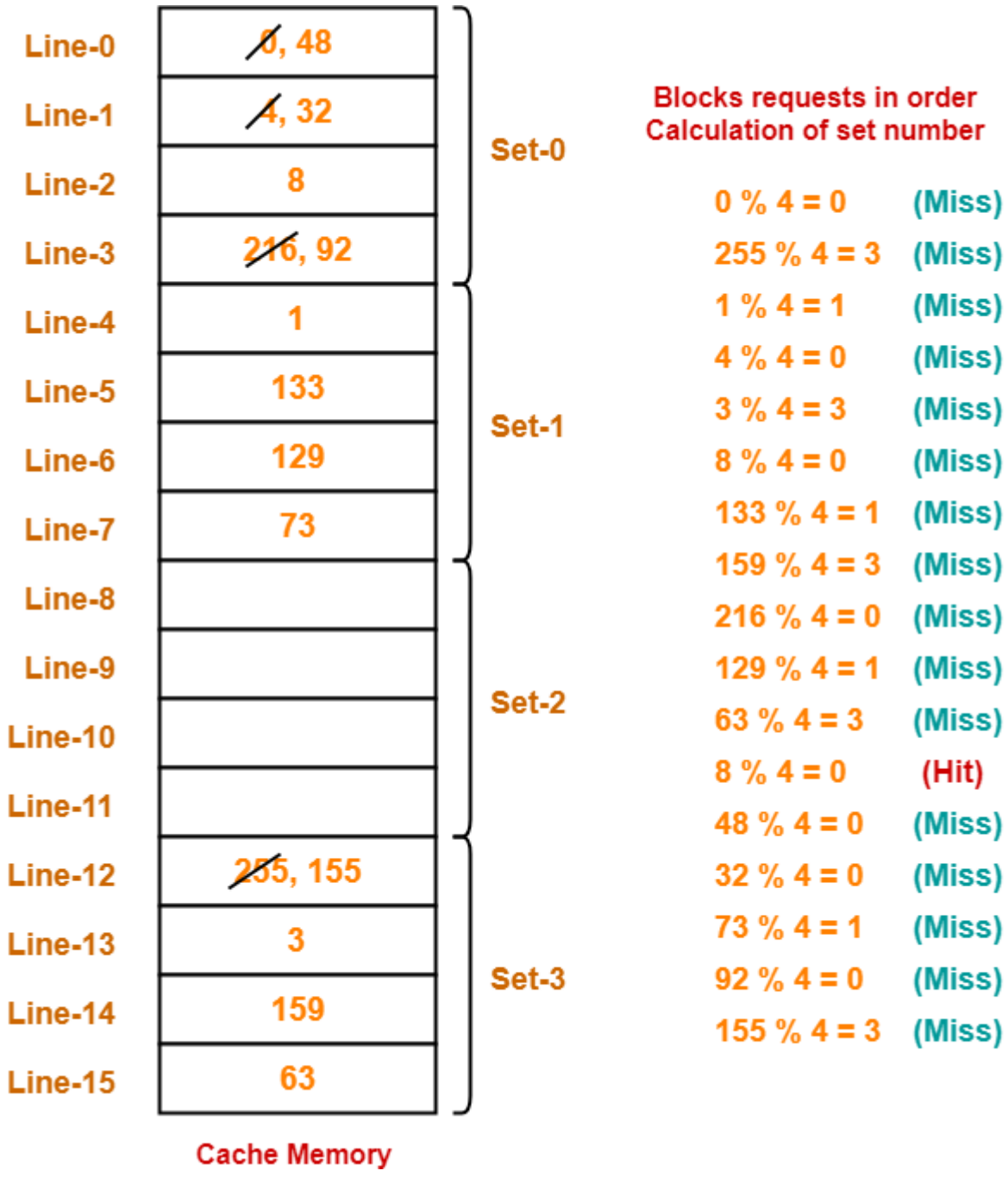
We have,

- There are 16 blocks in cache memory numbered from 0 to 15.
- Each set contains 4 cache lines.
- In set associative mapping, a particular block of main memory is mapped to a particular set of cache memory.
- The set number is given by-

Cache line number = Block address **modulo** Number of sets in cache

For the given sequence-

- Requests for memory blocks are generated one by one.
- The set number of the block is calculated using the above relation.
- Within that set, the block is placed in any freely available cache line.
- If all the blocks are already occupied, then one of the block is replaced in accordance with the employed replacement policy.




Thus,

- Out of given options, only block-216 is not present in the main memory.
- Option-(D) is correct.
- Hit ratio = 1 / 17
- Miss ratio = 16 / 17

**Also Read- Practice Problems On Set Associative Mapping**

# Problem-10:

Consider a small 2-way set associative mapping with a total of 4 blocks. LRU replacement policy is used for choosing the block to be replaced. The number of cache misses for the following sequence of block addresses 8, 12, 0, 12, 8 is ____.

# Solution-

- Practice yourself.
- Total number of cache misses = 4

# Problem-11:

Consider the cache has 4 blocks. For the memory references-

5, 12, 13, 17, 4, 12, 13, 17, 2, 13, 19, 13, 43, 61, 19

What is the hit ratio for the following cache replacement algorithms-

1. FIFO
2. LRU
3. Direct mapping
4. 2-way set associative mapping using LRU

# Solution-

- Practice yourself.
- Using FIFO as cache replacement algorithm, hit ratio = 5/15 = 1/3.
- Using LRU as cache replacement algorithm, hit ratio = 6/15 = 2/5.
- Using direct mapping as cache replacement algorithm, hit ratio = 1/15.
- Using 2-way set associative mapping as cache replacement algorithm, hit ratio = 5/15 = 1/3

# Problem-12:

A hierarchical memory system has the following specification, 20 MB main storage with access time of 300 ns, 256 bytes cache with access time of 50 ns, word size 4 bytes, page size 8 words. What will be the hit ratio if the page address trace of a program has the pattern 0, 1, 2, 3, 0, 1, 2, 4 following LRU page replacement technique?

# Solution-

Given-

- Main memory size = 20 MB
- Main memory access time = 300 ns
- Cache memory size = 256 bytes
- Cache memory access time = 50 ns
- Word size = 4 bytes
- Page size = 8 words

## Line Size-

We have,

Line size

= 8 words

= 8 x 4 bytes

= 32 bytes

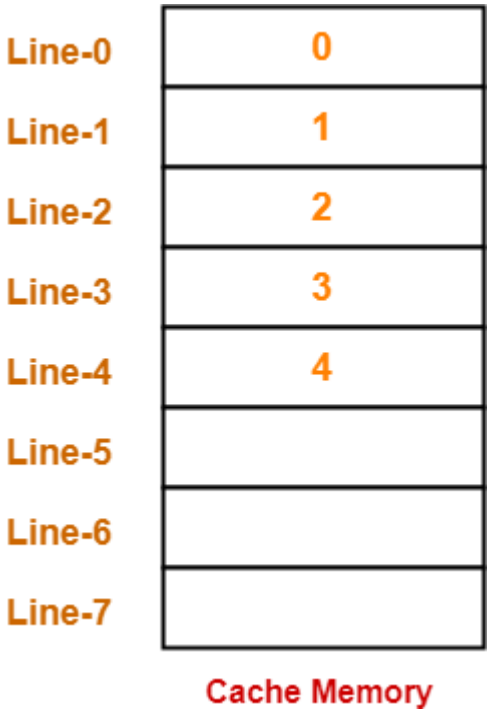
## Number of Lines in Cache-

Number of lines in cache

= Cache size / Line size

= 256 bytes / 32 bytes

= 8 lines




Thus,

- Number of hits = 3
- Hit ratio = 3 / 8

## Problem-13:

Consider an array A[100] and each element occupies 4 words. A 32 word cache is used and divided into 8 word blocks. What is the hit ratio for the following code-

```
for (i=0 ; i < 100 ; i++)
    A[i] = A[i] + 10;
```

## Solution-

Number of lines in cache

= Cache size / Line size

= 32 words / 8 words

= 4 lines

Since each element of the array occupies 4 words, so-

Number of elements that can be placed in one line = 2

Now, let us analyze the statement-

A[i] = A[i] + 10;

For each i,

- Firstly, the value of A[i] is read.
- Secondly, 10 is added to the A[i].
- Thirdly, the updated value of A[i] is written back.

Thus,

- For each i, A[i] is accessed two times.
- Two memory accesses are required-one for read operation and other for write operation.

Assume the cache is all empty initially.

In the first loop iteration,

- First, value of A[0] has to be read.
- Since cache is empty, so A[0] is brought in the cache.
- Along with A[0], element A[1] also enters the cache since each block can hold 2 elements of the array.


Cache Memory

- Thus, For A[0], a miss occurred for the read operation.
- Now, 10 is added to the value of A[0].

- Now, the updated value has to be written back to A[0].
- Again cache memory is accessed to get A[0] for writing its updated value.
- This time, for A[0], a hit occurs for the write operation.

In the second loop iteration,

- First, value of A[1] has to be read.
- A[1] is already present in the cache.
- Thus, For A[1]. a hit occurs for the read operation.
- Now, 10 is added to the value of A[1].
- Now, the updated value has to be written back to A[1].
- Again cache memory is accessed to get A[1] for writing its updated value.
- Again, for A[1], a hit occurs for the write operation.

In the similar manner, for every next two consecutive elements-

- There will be a miss for the read operation for the first element.
- There will be a hit for the write operation for the first element.
- There will be a hit for both read and write operations for the second element.

Likewise, for 100 elements, we will have 50 such pairs in cache and in every pair, there will be one miss and three hits.

Thus,

- Total number of hits = 50 x 3 = 150
- Total number of misses = 50 x 1 = 50
- Total number of references = 200 (100 for read and 100 for write)

Thus,

- Hit ratio = 150 / 200 = 3 / 4 = 0.75
- Miss ratio = 50 / 200 = 1 / 4 = 0.25

# Problem-14:

Consider an array has 100 elements and each element occupies 4 words. A 32 bit word cache is used and divided into a block of 8 words.

What is the hit ratio for the following statement-

```
for (i=0 ; i < 10 ; i++)
```

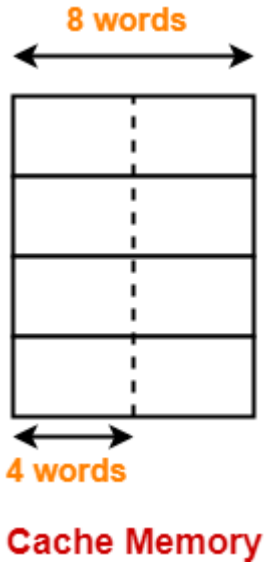
```
for (j=0 ; j < 10 ; j++)
A[i][j] = A[i][j] + 10;
```

if-

1. Row major order is used
2. Column major order is used

# Solution-

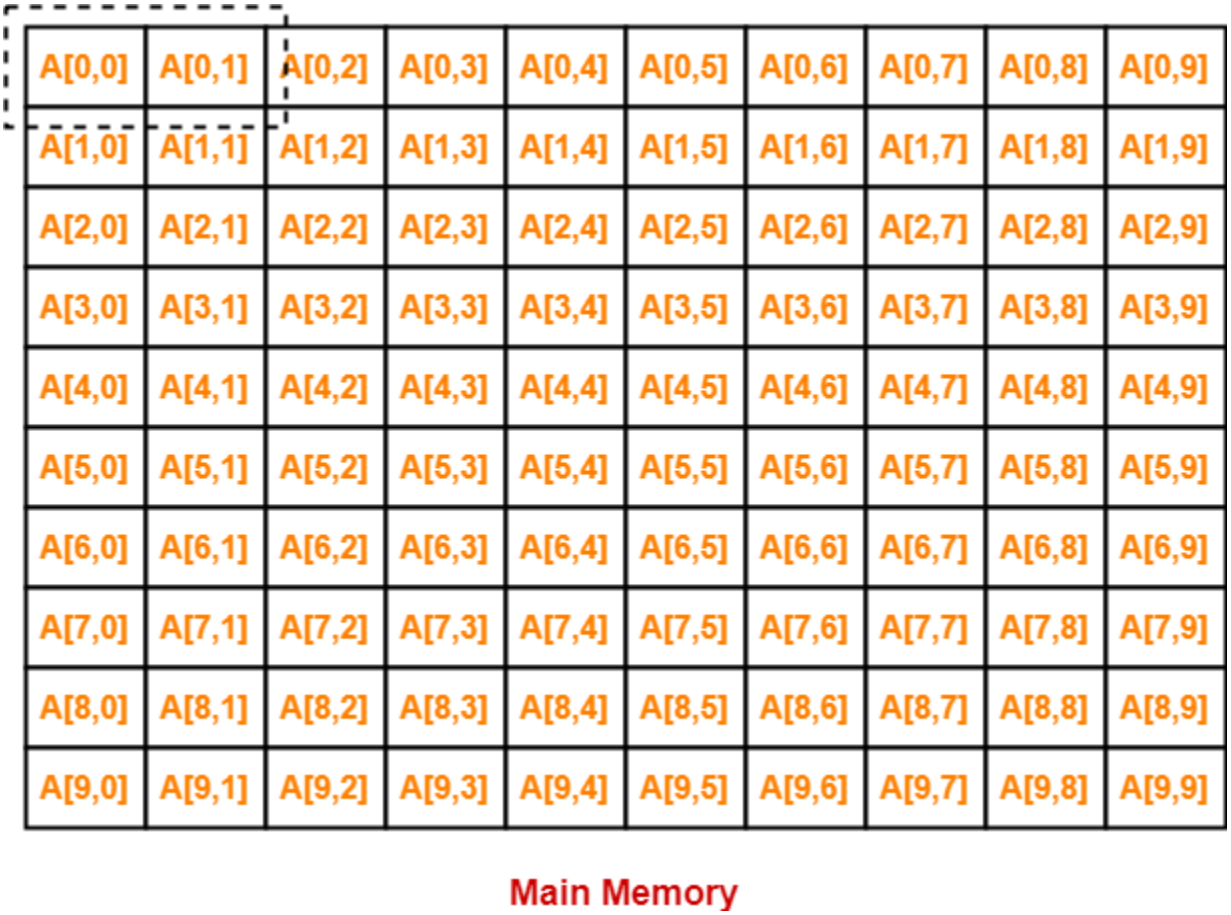
According to question, cache is divided as-




In each cache line, two elements of the array can reside.

## Case-01: When Row Major Order is Used-

In row major order, elements of the array are stored row wise in the memory as-




- In the first iteration, when A[0][0] will be brought in the cache, A[0][1] will also be brought.
- Then, things goes as it went in the previous question.
- There will be a miss for A[0,0] read operation and hit for A[0,0] write operation.
- For A[0,1], there will be a hit for both read and write operations.
- Similar will be the story with every next two consecutive elements.

Thus,

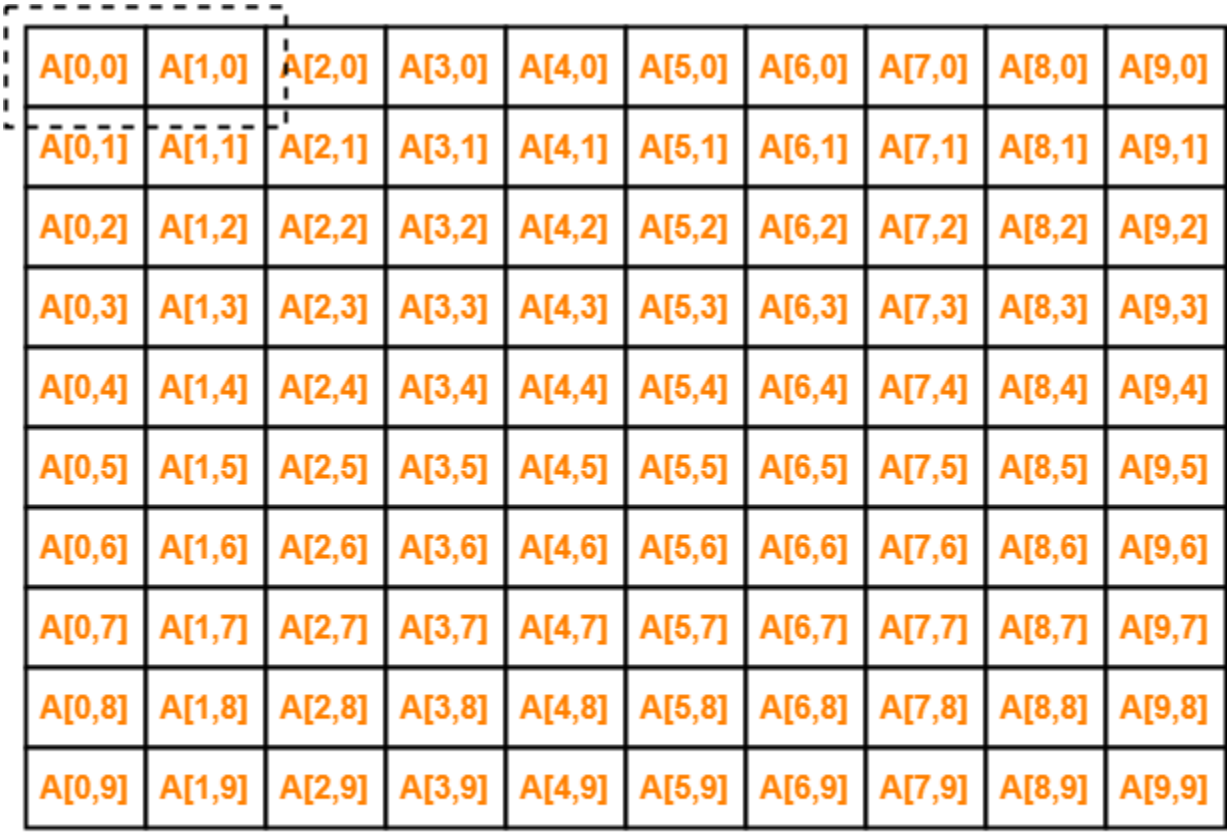
- Total number of hits = 50 x 3 = 150
- Total number of misses = 50 x 1 = 50
- Total number of references = 200 (100 for read and 100 for write)

Thus,

- Hit ratio = 150 / 200 = 3 / 4 = 0.75
- Miss ratio = 50 / 200 = 1 / 4 = 0.25

## Case-02: When Column Major Order is Used-

In column major order, elements of the array are stored column wise in the memory as-




**Main Memory**

- In the first iteration, when A[0,0] will be brought in the cache, A[1][0] will also be brought.
- This time A[1][0] will not be useful in iteration-2.
- A[1][0] will be needed after surpassing 10 element.
- But by that time, this block would have got replaced since there are only 4 lines in the cache.
- For A[1][0] to be useful, there has to be at least 10 lines in the cache.

Thus,

Under given scenario, for each element of the array-

- There will be a miss for the read operation.

- There will be a hit for the write operation.

Thus,

- Total number of hits = 100 x 1 = 100
- Total number of misses = 100 x 1 = 100
- Total number of references = 200 (100 for read and 100 for write)

Thus,

- Hit ratio = 100 / 200 = 1 / 2 = 0.50
- Miss ratio = 100 / 200 = 1 / 2 = 0.50

To increase the hit rate, following measures can be taken-

- Row major order can be used (Hit rate = 75%)
- The statement A[i][j] = A[i][j] + 10 can be replaced with A[j,i] = A[j][i] + 10 (Hit rate = 75%)

# Problem-15:

An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above by k. What is the miss ratio if the access sequence is passed through a cache of associativity A>=k exercising LRU replacement policy?

1. n/N
2. 1/N
3. 1/A
4. k/n

# Solution-

Required miss ratio = n/N.

Thus, Option (A) is correct

Q1.

Assume:
- A processor has a direct mapped cache
- Data words are *8 bits* long (i.e. 1 byte)
- Data addresses are *to the word*
- A physical address is *20 bits* long
- The tag is *11 bits*
- Each block holds 16 *bytes* of data

**How many blocks are in this cache?**

**Solution:**

Given that the physical address is 20 bits long, and the tag is 11 bits, there are 9 bits left over for the index and offset

We can determine the number of bits of offset as the problem states that:
- Data is word addressable and words are 8 bits long
- Each block holds 16 bytes

As there are 8 bits / byte, each block holds 16 words, thus **4** bits of **offset** are needed.

This means that there are 5 bits left for the index**. Thus, there are $2_5$ or 32 blocks in the cache**

**Q2.**

Consider a *16-way set-associative* cache
- Data words are *64 bits* long
- Words are addressed to the *half-word*
- The cache holds 2 Mbytes of data
- Each block holds 16 data words
- Physical addresses are 64 bits long

**How many bits of tag, index, and offset are needed to support references to this cache?**

Solution:

We can calculate the number of bits for the **offset** first:
- There are 16 data words per block which implies that at least 4 bits are needed
- Because data is addressable to the ½ word, an additional bit of offset is needed
- Thus, the **offset** is **5 bits**

To calculate the **index**, we need to use the information given regarding the total capacity of the cache:

- 2 MB is equal to $2^{21}$ total bytes.
- We can use this information to determine the total number of *blocks* in the cache…
o $2^{21}$ bytes **x** (1 block / 16 words) **x** (1 word / 64 bits) **x** (8 bits / 1 byte) = $2^{14}$ blocks
- Now, there are 16 (or $2^4$) blocks / set
o Therefore there are $2^{14}$ blocks **x** (1 set / $2^4$ blocks) = $2^{10}$ or 1024 sets
- Thus, **10 bits of index** are needed
Finally, the remaining bits form the tag:
- 64 – 5 – 10 = 49
- Thus, there are **49 bits of tag**
To summarize: **Tag**: 49 bits; **Index**: 10 bits; **Offset**: 5 bits

Q.3

• (a) A 64KB, direct mapped cache has 16 byte blocks. If addresses are 32 bits, how many bits are used the tag, index, and offset in this cache?
o Tag: 16 bits. Index: 12 bits. Offset: 4 bits.
o In a 64KB direct mapped cache with 16 byte blocks there are 4096 cache lines (blocks), therefore requiring 12 bits for the index (i.e., to select the block). Since the blocks are 16 bytes each, 4 bits are required for the offset. The remaining 16 bits are used for the tag. Note that the tag bits (for direct mapped) can also be derived as total address bits (32) minus total bits to address the cache size (i.e., to address 64KB we need 16 bits).
• (b) How would the address be divided if the cache were 4-way set associative instead?
o Tag: 18 bits. Index: 10 bits. Offset: 4 bits. If the cache were 4-way associative there would be 4 blocks in each set and 1024 sets or lines. Therefore we require 10 bits for the index (to specify the set to where the block is mapped). The offset would remain at 4 bits, and the tag would now use 18 bits.
• (c) How many bits is the index for a fully associative cache. Explain your answer.
o A fully associative cache has no bits in the index, since any address can be stored anywhere in the cache.